

Inner Product Spaces

Math Foundations — Node 08

First, an example

Before any axioms, look at the standard dot product on $\mathbb{R}^2$. Take $u = (3, 4)$ and $v = (-4, 3)$. Then

$$\langle u, v \rangle = 3 \cdot (-4) + 4 \cdot 3 = -12 + 12 = 0,$$

Read aloud: “inner product of u and v equals three times negative four plus four times three, which is zero.” so u and v are *orthogonal*. Their lengths are

$$\|u\| = \sqrt{3^2 + 4^2} = 5, \quad \|v\| = \sqrt{(-4)^2 + 3^2} = 5.$$

Read aloud: “norm of u equals the square root of nine plus sixteen, which is five; likewise norm of v equals five.” So u and v are two perpendicular vectors of the same length 5. The whole chapter is about which abstract structures preserve this “angle and length” picture.

1 Motivation

In $\mathbb{R}^n$ the dot product $u \cdot v = \sum_i u_i v_i$ encodes both length, via $\|u\|^2 = u \cdot u$, and angle, via the relation $\cos \theta = (u \cdot v) / (\|u\| \|v\|)$. An *inner product space* is the abstraction that lets us carry these geometric notions into any vector space, including spaces of functions, sequences, and random variables. Linear algebra without an inner product gives us linear maps; with one, it gives us geometry.

2 Definitions

Let V be a vector space over $\mathbb{F} \in \{\mathbb{R}, \mathbb{C}\}$.

Definition 1 (Inner product). An *inner product* on V is a map $\langle \cdot, \cdot \rangle : V \times V \rightarrow \mathbb{F}$ satisfying, for all $u, v, w \in V$ and $\alpha \in \mathbb{F}$:

1. **Conjugate symmetry:** $\langle u, v \rangle = \overline{\langle v, u \rangle}$.
2. **Linearity in the first argument:** $\langle \alpha u + w, v \rangle = \alpha \langle u, v \rangle + \langle w, v \rangle$.
3. **Positive definiteness:** $\langle u, u \rangle \geq 0$, with equality iff $u = 0$.

A vector space equipped with an inner product is called an *inner product space*.

Note that over $\mathbb{R}$, conjugate symmetry collapses to symmetry, and linearity in the first argument together with symmetry gives linearity in the second argument as well. Over $\mathbb{C}$, the second argument is *conjugate*-linear; this is sometimes called the “physicist’s convention” when the conjugate sits in the first slot — we follow the mathematician’s convention here.

Definition 2 (Induced norm). The *induced norm* on an inner product space $(V, \langle \cdot, \cdot \rangle)$ is

$$\|v\| := \sqrt{\langle v, v \rangle}, \quad v \in V.$$

Read aloud: “norm of v is defined as the square root of the inner product of v with itself, for every v in V .” Positive definiteness makes this well-defined and non-negative.

Definition 3 (Orthonormal set). A finite or countable family $\{e_i\}_{i \in I} \subset V$ is called *orthonormal* if

$$\langle e_i, e_j \rangle = \delta_{ij} = \begin{cases} 1 & i = j, \\ 0 & i \neq j. \end{cases}$$

Read aloud: “inner product of e_i and e_j equals the Kronecker delta δ_{ij} , which is one when i equals j and zero otherwise.”

3 The Cauchy-Schwarz inequality

Theorem 4 (Cauchy-Schwarz). Let $(V, \langle \cdot, \cdot \rangle)$ be an inner product space over $\mathbb{F}$. For all $u, v \in V$,

$$|\langle u, v \rangle| \leq \|u\| \|v\|, \quad (1)$$

Read aloud: “the absolute value of the inner product of u and v is less than or equal to the norm of u times the norm of v .” with equality if and only if u and v are linearly dependent.

Proof. If $v = 0$ both sides of (1) are zero and there is nothing to prove, so assume $v \neq 0$. The trick is to consider, for a scalar $t \in \mathbb{F}$ to be chosen, the non-negative quantity

$$0 \leq \|u - tv\|^2 = \langle u - tv, u - tv \rangle.$$

Read aloud: “zero is less than or equal to the squared norm of u minus t times v , which equals the inner product of that vector with itself.” Expanding using linearity in the first slot and conjugate linearity in the second,

$$\begin{aligned} \langle u - tv, u - tv \rangle &= \langle u, u \rangle - t \langle v, u \rangle - \bar{t} \langle u, v \rangle + |t|^2 \langle v, v \rangle \\ &= \|u\|^2 - t \overline{\langle u, v \rangle} - \bar{t} \langle u, v \rangle + |t|^2 \|v\|^2. \end{aligned}$$

Read aloud: “the inner product of u minus tv with itself expands as the inner product of u with u , minus t times the inner product of v with u , minus t -bar times the inner product of u with v , plus the modulus of t squared times the inner product of v with v ; equivalently, norm- u squared minus t times the conjugate of the inner product of u and v , minus t -bar times the inner product of u and v , plus the modulus of t squared times norm- v squared.” Choose $t = \langle u, v \rangle / \|v\|^2$. Then $\bar{t} \langle u, v \rangle = |\langle u, v \rangle|^2 / \|v\|^2$ and $t \overline{\langle u, v \rangle} = |\langle u, v \rangle|^2 / \|v\|^2$, while $|t|^2 \|v\|^2 = |\langle u, v \rangle|^2 / \|v\|^2$. Substituting,

$$0 \leq \|u\|^2 - \frac{|\langle u, v \rangle|^2}{\|v\|^2}.$$

Read aloud: “zero is less than or equal to norm- u squared minus the modulus squared of the inner product of u and v , divided by norm- v squared.” Multiplying through by $\|v\|^2 > 0$ and rearranging,

$$|\langle u, v \rangle|^2 \leq \|u\|^2 \|v\|^2,$$

Read aloud: “the modulus squared of the inner product of u and v is less than or equal to norm- u squared times norm- v squared.” and taking square roots gives (1).

For the equality case: equality forces $\|u - tv\|^2 = 0$, so by positive definiteness $u = tv$, i.e. u and v are linearly dependent. Conversely if $u = \lambda v$ then both sides of (1) equal $|\lambda| \|v\|^2$. $\square$

Proposition 5 (Triangle inequality). *For all $u, v \in V$, $\|u + v\| \leq \|u\| + \|v\|$.*

Proof. Expand $\|u + v\|^2 = \|u\|^2 + 2\Re\langle u, v \rangle + \|v\|^2$, bound $\Re\langle u, v \rangle \leq |\langle u, v \rangle| \leq \|u\| \|v\|$ by Theorem 4, and take square roots. $\square$

4 Examples

Example 6 (Euclidean space). $V = \mathbb{R}^n$, $\langle u, v \rangle = \sum_{i=1}^n u_i v_i$. The induced norm is the usual Euclidean length, and Cauchy-Schwarz is the classical $(\sum u_i v_i)^2 \leq (\sum u_i^2)(\sum v_i^2)$.

Example 7 (L^2 on the unit interval). $V = C[0, 1]$ with $\langle f, g \rangle = \int_0^1 f(x) g(x) dx$. Cauchy-Schwarz becomes

$$\left| \int_0^1 f g dx \right| \leq \left(\int_0^1 f^2 dx \right)^{1/2} \left(\int_0^1 g^2 dx \right)^{1/2},$$

Read aloud: “the absolute value of the integral from zero to one of f times g dx is less than or equal to the square root of the integral of f squared, times the square root of the integral of g squared.” a special case of Hölder’s inequality with $p = q = 2$.

5 Where this leads

Cauchy-Schwarz is the inequality that lets us define $\cos \theta = \langle u, v \rangle / (\|u\| \|v\|) \in [-1, 1]$. The construction of orthonormal bases via Gram-Schmidt, and orthogonal projection onto a closed subspace, both rest on this single inequality. In ML, every similarity score that takes the form of a dot product — attention logits, contrastive loss similarities, cosine similarity — is a direct corollary.